

tf.compat.v1.nn.softmax_cross_entropy_with_logits

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/softmax_cross_entropy_with_logits

Computes softmax cross entropy between logits and labels. (deprecated)

Function Signatures

```
tf.compat.v1.nn.softmax_cross_entropy_with_logits(  
    labels=None, logits=None, dim=-1, name=None, axis=None )
```

Code Examples

```
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Documentation

Computes softmax cross entropy between logits and labels. (deprecated)

Future major versions of TensorFlow will allow gradients to flow into the labels input on backprop by default.

See `tf.nn.softmax_cross_entropy_with_logits_v2`.

Measures the probability error in discrete classification tasks in which the classes are mutually exclusive (each entry is in exactly one class). For example, each CIFAR-10 image is labeled with one and only one label: an image can be a dog or a truck, but not both.

If using exclusive labels (wherein one and only one class is true at a time), see `sparse_softmax_cross_entropy_with_logits`.

A common use case is to have logits and labels of shape `[batch_size, num_classes]`, but higher dimensions are supported, with the `dim` argument specifying the class dimension.

Backpropagation will happen only into logits. To calculate a cross entropy loss that allows backpropagation into both logits and labels, see `tf.nn.softmax_cross_entropy_with_logits_v2`.

Note that to avoid confusion, it is required to pass only named arguments to this function.

Returns

